



Course Information

Course Code: CS-11303

Credit Hours: 4 (3,1)

Lab Hours/Week: 3

Course Title: **Operating System**

Lecture Hours/Week: 3

Pre-Requisites: **None**

Teaching Tools and Technologies

To support conceptual understanding of Operating Systems theory multiple visualization, simulation, and analytical tools are incorporated.

1. Process & CPU Scheduling Visualization Tools

To demonstrate process lifecycle, context switching, and scheduling:

- CPU Scheduling simulators (FCFS, SJF, Priority, Round Robin)
- Gantt Chart Generators
- Process State Transition diagrams
- Ready Queue visualization tools

These tools help students understand waiting time, turnaround time, response time, and scheduling efficiency.

2. Memory Management Demonstrators

To reinforce paging, segmentation, and virtual memory concepts:

- Paging address translation visualizers
- TLB lookup simulation diagrams
- Page replacement animation tools (FIFO, LRU, Optimal)
- Fragmentation calculators

These tools allow students to see logical-to-physical mapping clearly.

3. Deadlock & Synchronization Modeling Tools

- Resource Allocation Graph visualizers
- Banker's Algorithm step simulation sheets
- Critical Section animations
- Semaphore and mutex conceptual demonstrations

Students observe deadlock detection and avoidance in controlled scenarios.

4. File System & Storage Conceptual Models

- File allocation technique diagrams (Contiguous, Linked, Indexed)
- Disk scheduling animation tools (FCFS, SSTF, SCAN, C-SCAN)
- Inode structure models (Linux-based reference)
- Directory tree modeling diagrams

5. Virtualization & Modern OS Concepts

- Virtual machine architecture diagrams
- Hypervisor-based system models
- Container vs VM conceptual comparison
- Cloud OS infrastructure discussion modules

Integration in Student Learning (Theory)

These tools align from CLO-1 to CLO-7 and ensure:

- Analytical understanding of OS design
- Algorithmic reasoning ability
- System-level thinking
- Preparation for Linux administration and cybersecurity careers

Theoretical rigor ensures compliance with HEC curriculum standards and prepares students for advanced system-level learning.